

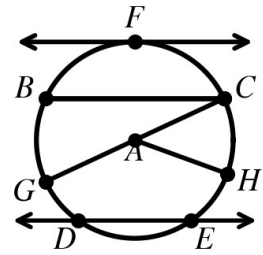
Speak Like A Mathematician:

circle:

chord:

secant:

tangent:



Tangent Facts

A tangent line is perpendicular to the radius drawn to the point of tangency.

Two tangent segments drawn to a circle from the SAME exterior point are _____ .

Example 1 - Name the Parts of Circle A

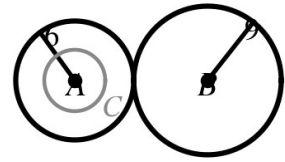
Use the figure at the top right.

- Name a chord that is NOT a diameter.
- Name a secant.
- Name the tangent and its point of tangency.
- Name a diameter.
- Name a radius that is not part of a diameter.

Example 2 - Radii and Diameters

Circle A has radius 6 and circle B has radius 9.

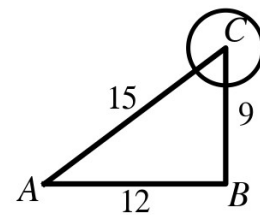
- Find the diameter of each circle.
- Are circles A and B congruent? Are they similar?



Example 3 - Is It Tangent?

In the triangle at the right, $AB = 12$, $BC = 9$, and $AC = 15$.

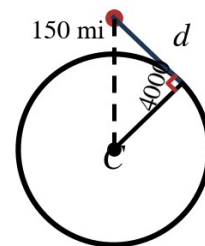
- Is AB tangent to circle C ? Justify with numbers.
- What would your answer be if AC were 16?



Example 4 - How Far Can the Satellite See?

A satellite orbits 150 mi above Earth (radius ≈ 4000 mi). Its line of sight d is tangent to Earth.

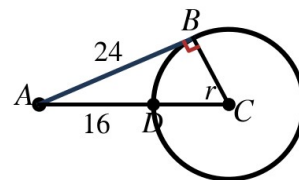
- Explain why the triangle formed is a right triangle.
- Find d to the nearest mile.



Example 5 - Find the Radius

In the figure, AB is tangent at B , $AB = 24$, and $AD = 16$.

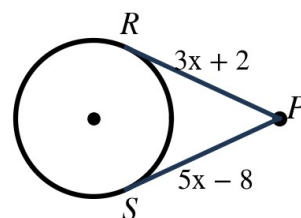
- Write an equation using the right triangle ABC .
- Solve for r .



Example 6 - Congruent Tangent Segments

PR and PS are tangent segments, $PR = 3x + 2$ and $PS = 5x - 8$.

- Find x .
- Find PR .



Example 7 - Counting Common Tangents

- Two circles are externally tangent (they touch at one point). How many common tangent lines do they have?
- One circle lies completely outside the other (no touching). How many?
- The circles are concentric. How many?